

Project Part Two

CS 421: Natural Language Processing

Due: May 1, 2026 (11:59 p.m. CST)

1 Introduction

Welcome to Project Part Two! In this part, you will explore the application of NLP techniques for the task of natural language generation. You will be working with the WASSA 2024 Track-2 dataset as in Project Part One, which contains multi-turn dialogues annotated with emotion intensity scores, emotion polarity scores, and empathy intensity scores. The dataset is based on participants' empathic reactions to news stories, where multiple readers read the same article and then engage in conversations about it. These dialogues capture how people express emotions and empathy when reacting to situations involving harm to a person, group, or community. The goal of this task is, given the desired emotion intensity, polarity, and empathy level, generate the next utterance in the conversation.

As always, if you encounter questions while working on this project, please post them on Piazza or bring them up during office hours. Good luck, and happy coding!

2 Instructions

Each part of this deliverable is labeled as Code or Written. The guidelines for these two types of components are provided below.

2.1 Code

All code must be completed using Python (version 3.10+). You are required to implement **two methods**:

- Corpus Based Chatbot
- In-Context Learning based methods for LLM Chatbots

For the training transformer-based methods, you may access the free Google Colab Pro for Education (US-based students + faculty) to use GPUs for training. While we recommend Google Colab, you may also access GPU resources through <https://lightning.ai>, www.kaggle.com, etc.

Reminder: Detailed instructions as well as tips for using Colab Pro for Education are given with Project Part One. Remember that each student has **100 CUs/month**. This is usually sufficient if you follow the tips above. Treat compute units like a budget: spend them wisely on final experiments, not on debugging.

Submission format: You are required to submit:

- ☐ `generations_corpus.csv, generations_icl.csv`: **Test output files** (csv) containing your model outputs
- ☐ Your **code files** (.py or .ipynb)
- ☐ A **detailed README** (.txt or .md file) including:
 - Implementation details for each method
 - Results obtained on the development set
 - Notes on preprocessing or model choices
 - Clear and easy to follow instructions on how to run your code

You must compress these files together as a **zip** file to be submitted on Gradescope under **Project Part Two**. All specified files need to be submitted to receive full credit.

2.2 Written

You are required to submit all Written questions in a single PDF file. You may create this PDF using Microsoft Word, scans of your handwritten solution, L^AT_EX or any other method or design tool you prefer. To submit your solution for Written questions, you need to provide answers to the following questions in a single PDF.

- ☐ Q3

Before submission, ensure that all pages of your solution are present and in order. Submit this PDF on Gradescope under **Project Part Two**.

3 Questions

3.1 Code (65 points)

For all following questions, you will utilize the WASSA 2024 Track-2 dataset provided in the assignment zip file on Blackboard. The zip file has dataset files for train, development and test sets. You may create any helper functions or files or loading and processing of the data. You are free to use any functions or files created in Part One of the project. However, these must be submitted on Gradescope.

Q1 (35): Corpus-Based Chatbot

In this part, you will be implementing a corpus-based chatbot using the utterances provided in the `text` column of dataset files. You will perform the following steps:

1. Load the data as in Part One of the project.
2. For each sample in the training data file, find the contextual embeddings of the utterances using the SentenceTransformer¹ model.
3. Normalize the emotion intensity and empathy scores to be in the range $[0, 1]$ and convert the emotion polarity label into a one-hot encoded vector.
4. Calculate the similarity score between the training utterances and the query utterance, S , as:

$$s_{\text{text}} = \cos(\mathbf{e}_1, \mathbf{e}_2) \quad (1)$$

$$s_{\text{emotion}} = 1 - |\text{EI}_1 - \text{EI}_2| \quad (2)$$

$$s_{\text{empathy}} = 1 - |\text{Emp}_1 - \text{Emp}_2| \quad (3)$$

$$s_{\text{polarity}} = \begin{cases} 1, & \text{if } P_1 = P_2 \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

$$S_{\text{total}} = w_1 s_{\text{text}} + w_2 s_{\text{emotion}} + w_3 s_{\text{empathy}} + w_4 s_{\text{polarity}} \quad (5)$$

Here, $w_1, \dots, w_4$ are tunable weights for each component of similarity. Choose appropriate weights and mention your choice as well as a reflection with all of your tested choices in the report.

The query sentence is a concatenation of the utterances conversation history until the start of generation with the desired emotion intensity, empathy and polarity.

5. Find the most similar sentence in the training corpus using the calculated similarity scores and output that as the predicted next sentence.
6. Follow these steps till you achieve the desired number of utterances to be generated.

Follow these steps for generating the next 5 conversation turns starting at turn 6 for all unique `conversation_id` in the validation and test set.

You will use the following automated metrics: ROGUE-1,2&L², BLEU³, and BertScore⁴. Compute and report the automated metrics on the validation set and submit a csv file of your generations for the test set. Generate 5 chat conversations with at least 10 generated utterances in each conversation for analysis.

¹https://sbert.net/docs/package_reference/sentence_transformer/SentenceTransformer.html

²<https://huggingface.co/spaces/evaluate-metric/rouge>

³<https://huggingface.co/spaces/evaluate-metric/bleu>

⁴<https://huggingface.co/spaces/evaluate-metric/bertscore>

Q2 (30): In-context learning with Large Language Models

In this question, you will explore in-context learning (ICL) using a pretrained Large Language Model (e.g., **Llama**⁵,⁶, or **Phi**⁷) via the HuggingFace Transformers library. For ICL, you will design prompts that include a few-shot set of example conversations (input + gold response) directly within the input context to guide the model's generation. In this task, each input will provide the conversation history up to **turn 5**, and your model must generate the next **10 utterances**, starting from **turn 6**.

You will:

- Load a pretrained causal or instruction-tuned transformer model and its tokenizer from HuggingFace.⁸
- Construct few-shot ICL prompts by concatenating a set of labeled examples (input + gold output) followed by an unlabeled conversation instance for the specific turn.
- Experiment with at least two few-shot configurations⁹, such as **3-shot** and **5-shot**, on a small subset of the development data (e.g, 30-50 instances) to assess which setting yields better responses, as model performance often varies with the number of examples.
- Evaluate the LLM on the development dataset and report performance using standard metrics (ROGUE-1,2&L, BLEU, and BertScore).
- Generate a conversation response for every sample provided in the test dataset and save them in a CSV file for submission. The CSV file should contain 3 columns - `id`, `turn_number`, and `generated_response`, and each row should contain the predicted value for the corresponding conversation ID.

3.2 Written (35 points)

Q3 (35): Qualitative Evaluation of conversations

Select five conversations from the development dataset. For each conversation, analyze the responses generated from turn 6 through turn 10 and compare the dialogues produced across the two modeling approaches. For each turn (6-10) in every selected conversation, evaluate the generated responses from both approaches.

Use the best model from your Part 1 approach to obtain the emotion polarity, intensity, and empathy scores for the generated responses. You will then

⁵Llama 3.2: <https://huggingface.co/meta-llama/Llama-3.2-1B-Instruct>

⁶Llama 3.1: <https://huggingface.co/meta-llama/Llama-3.1-8B-Instruct>

⁷Phi-3-mini: <https://huggingface.co/microsoft/Phi-3-mini-4k-instruct>

⁸Transformer: <https://huggingface.co/docs/transformers/en/quicktour>

⁹Prompting: <https://huggingface.co/docs/transformers/v4.39.2/tasks/prompting>

analyze these conversations for other generation aspects based on the following questions, to be answered on a scale of 1–5 (Likert scale), where 1 means 'Strongly Disagree', 2 means 'Disagree', 3 means 'Neither agree or disagree', 4 means 'Agree' and 5 means 'Strongly Agree')

1. Were the responses generated fluent? That is, were they grammatically well-formed and fluent in the English language?
2. Were the responses generated relevant? That is, in the context of the user input, did the model provide a relevant response?
3. Were the responses generated coherent? That is, do the model provide consistent responses to user input given the previous utterances in the conversation?
4. Were the scores for emotional polarity, intensity, and empathy of the generated responses aligned with the expected values based on the input?
5. Lastly, overall, which approach worked the best and why? Support with examples.

You should present your results clearly in your written report (PDF), for example, using tables or bullet points, and provide a brief explanation supported by examples for each of your answers.

Example of qualitative evaluation on a single utterance

last_utterance: I just lost my job and feel completely hopeless.
 Gold (Expected): Polarity = Pos, Intensity = 4, Empathy = 5

Corpus Model: I'm sorry to hear that. Losing a job can be really hard, but things can get better with time.
 Predicted through Part-1: Polarity = Pos, Intensity = 3, Empathy = 3

ICL Model: That must feel really difficult. You've been through a lot! Want to talk about what happened?
 Predicted through Part-1: Polarity = Pos, Intensity = 4, Empathy = 5

Aspect	Corpus-based	ICL
Fluency	5	5
Relevance	4	3
Coherence	4	4
Emotion Alignment (vs. Gold)	3	5

4 Rubric

This assignment will be graded according to the rubric below. Partial points may be awarded for rubric items at the discretion of the course staff.

Q1 (35 points possible)	+35
Q2 (30 points possible)	+30
Q3 (35 points possible)	+35